ANOVA Calculations — Code Explanation Quality Study

Significance level: = 0.20 (80% confidence)
Scale: 0–20 | N = 180 observations

Formulas Reference

Formula	Expression
Sample Mean	$\bar{X} = \sum X_i/n$
Sample Variance	$s^2 = \sum(X_i - \bar{X})^2/(n-1)$
Standard Deviation	$s = \sqrt{s^2}$
Grand Mean	$GM = \sum n_i \bar{x}_i / \sum n_i$
SS(B)	$\sum n_i(\bar{X}_i - GM)^2$
SS(W)	$\sum df_i \cdot s_i^2$
SS(T)	$SS(B) + SS(W)$
MS(B)	$SS(B)/(k-1)$
MS(W)	$SS(W)/(n-k)$
F	$MS(B)/MS(W)$
SS(AxB)	$SS(M) - SS(A) - SS(B)$

H1 — One-Way ANOVA — Effect of Model

Step 1 — Sample Characteristics

Model	n	Mean ($\bar{X}$)	s	s^2
Claude Sonnet 4.6	60	17.533	0.650	0.423
DeepSeek V3	60	16.617	0.922	0.851
Gemini 3 Flash	60	14.850	0.777	0.604

k = 3, n = 60 per group, N = 180

Assumption check — ratio of largest to smallest standard deviation must be $< 2{:}1$:

$$\frac{s_{max}}{s_{min}} = \frac{0.922}{0.650} = 1.42 < 2\checkmark$$

Step 2 — Grand Mean

$$GM = \frac{\sum n_i \bar{x}_i}{\sum n_i} = \frac{60 \times 17.533 + 60 \times 16.617 + 60 \times 14.850}{180}$$

$$GM = \frac{1051.98 + 997.02 + 891.00}{180} = \frac{2940.00}{180} = \boxed{16.333}$$

Step 3 — SS(B) Between-Group Sum of Squares

$$SS(B) = \sum n_i (\bar{X}_i - GM)^2$$

- **Claude:** $60 \times (17.533 - 16.333)^2 = 60 \times (1.200)^2 = 60 \times 1.440 = 86.400$
- **DeepSeek:** $60 \times (16.617 - 16.333)^2 = 60 \times (0.284)^2 = 60 \times 0.081 = 4.840$
- **Gemini:** $60 \times (14.850 - 16.333)^2 = 60 \times (-1.483)^2 = 60 \times 2.199 = 131.993$

$$SS(B) = 86.400 + 4.840 + 131.993 = \boxed{223.233}$$

Step 4 — SS(W) Within-Group Sum of Squares

$$SS(W) = \sum df_i \cdot s_i^2$$

- **Claude:** $(60 - 1) \times 0.423 = 59 \times 0.423 = 24.957$
- **DeepSeek:** $(60 - 1) \times 0.851 = 59 \times 0.851 = 50.209$
- **Gemini:** $(60 - 1) \times 0.604 = 59 \times 0.604 = 35.636$

$$SS(W) = 24.957 + 50.209 + 35.636 = \boxed{110.802}$$

Verification: $SS(T) = 223.233 + 110.802 = 334.035 \approx 333.999$

Step 5 — MS and F

$$MS(B) = \frac{SS(B)}{k-1} = \frac{223.233}{2} = 111.617$$

$$MS(W) = \frac{SS(W)}{N-k} = \frac{110.802}{177} = 0.626$$

$$F = \frac{MS(B)}{MS(W)} = \frac{111.617}{0.626} = \boxed{178.358}$$

Step 6 — ANOVA Table

	SS	df	MS	F	P
Between	223.233	k−1 = 2	223.233/2 = 111.617	111.617/0.626 = 178.358	Tail area above F = **< 0.001**
Within	110.767	n−k = 177	110.767/177 = 0.626		
Total	333.999	n−1 = 179			

Step 7 — Decision

- **F_crit** (= 0.20, df = 2, 177) = **1.624**
- **p-value = < 0.001**

$$F = 178.358 \gg F_{crit} = 1.624 \quad \text{and} \quad p < 0.001 < \alpha = 0.20$$

→ H0 is Rejected.

Post-hoc Tukey HSD confirms all pairwise comparisons significant:
Claude (17.53) > DeepSeek (16.62) > Gemini (14.85)

H2 — One-Way ANOVA — Effect of Programming Language

Step 1 — Sample Characteristics

Language	n	Mean (X̄)	s	s^2
Python	90	16.511	1.309	1.713
C++	90	16.156	1.406	1.976

k = 2, n = 90 per group, N = 180

Assumption check:

$$\frac{s_{max}}{s_{min}} = \frac{1.406}{1.309} = 1.07 < 2\checkmark$$

Step 2 — Grand Mean

$$GM = \frac{90 \times 16.511 + 90 \times 16.156}{180} = \frac{1485.99 + 1454.04}{180} = \frac{2940.03}{180} = \boxed{16.333}$$

Step 3 — SS(B)

$$SS(B) = \sum n_i(\bar{X}_i - GM)^2$$

- **Python:** $90 \times (16.511 - 16.333)^2 = 90 \times (0.178)^2 = 90 \times 0.0317 = 2.844$
- **C++:** $90 \times (16.156 - 16.333)^2 = 90 \times (-0.177)^2 = 90 \times 0.0314 = 2.844$

$$SS(B) = 2.844 + 2.844 = \boxed{5.689}$$

Step 4 — SS(W)

$$SS(W) = \sum df_i \cdot s_i^2$$

- **Python:** $89 \times 1.713 = 152.457$
- **C++:** $89 \times 1.976 = 175.864$

$$SS(W) = 152.457 + 175.864 = \boxed{328.321}$$

Verification: $SS(T) = 5.689 + 328.321 = 334.010 \approx 333.999$

Step 5 — MS and F

$$MS(B) = \frac{SS(B)}{k-1} = \frac{5.689}{1} = 5.689$$

$$MS(W) = \frac{SS(W)}{N-k} = \frac{328.321}{178} = 1.845$$

$$F = \frac{MS(B)}{MS(W)} = \frac{5.689}{1.845} = \boxed{3.084}$$

Step 6 — ANOVA Table

	SS	df	MS	F	P
Between	5.689	k−1 = 1	5.689/1 = 5.689	5.689/1.845 = 3.084	Tail area above F = **0.081**
Within	328.311	n−k = 178	328.311/178 = 1.845		
Total	333.999	n−1 = 179			

Step 7 — Decision

- **F__crit** (= 0.20, df = 1, 178) = **1.655**
- **p-value = 0.081**

$$F = 3.084 > F_{crit} = 1.655 \quad \text{and} \quad p = 0.081 < \alpha = 0.20$$

→ H0 is Rejected.
Models perform significantly better in Python (M = 16.51) than in C++ (M = 16.16).

H3 — One-Way ANOVA — Effect of Function Complexity

Step 1 — Sample Characteristics

Complexity	n	Mean (X̄)	s	s²
Easy	60	16.400	1.532	2.346
Medium	60	16.333	1.188	1.412
Hard	60	16.267	1.376	1.894

k = 3, n = 60 per group, N = 180

Assumption check:

$$\frac{s_{max}}{s_{min}} = \frac{1.532}{1.188} = 1.29 < 2\checkmark$$

Step 2 — Grand Mean

$$GM = \frac{60 \times 16.400 + 60 \times 16.333 + 60 \times 16.267}{180} = \frac{2940.00}{180} = \boxed{16.333}$$

Step 3 — SS(B)

$$SS(B) = \sum n_i(\bar{X}_i - GM)^2$$

- **Easy:** $60 \times (16.400 - 16.333)^2 = 60 \times (0.067)^2 = 60 \times 0.00449 = 0.267$
- **Medium:** $60 \times (16.333 - 16.333)^2 = 60 \times 0 = 0$
- **Hard:** $60 \times (16.267 - 16.333)^2 = 60 \times (-0.066)^2 = 60 \times 0.00436 = 0.267$

$$SS(B) = 0.267 + 0 + 0.267 = \boxed{0.533}$$

Step 4 — SS(W)

$$SS(W) = \sum df_i \cdot s_i^2$$

- **Easy:** $59 \times 2.346 = 138.414$
- **Medium:** $59 \times 1.412 = 83.308$
- **Hard:** $59 \times 1.894 = 111.746$

$$SS(W) = 138.414 + 83.308 + 111.746 = \boxed{333.467}$$

Verification: $SS(T) = 0.533 + 333.467 = 334.000 \approx 333.999$

Step 5 — MS and F

$$MS(B) = \frac{0.533}{2} = 0.267$$

$$MS(W) = \frac{333.467}{177} = 1.884$$

$$F = \frac{0.267}{1.884} = \boxed{0.142}$$

Step 6 — ANOVA Table

	SS	df	MS	F	P
Between	0.533	k−1 = 2	0.533/2 = 0.267	0.267/1.884 = 0.142	Tail area above F = **0.868**
Within	333.467	n−k = 177	333.467/177 = 1.884		
Total	333.999	n−1 = 179			

Step 7 — Decision

- **F_crit** (= 0.20, df = 2, 177) = **1.624**
- **p-value = 0.868**

$$F = 0.142 \ll F_{crit} = 1.624 \quad \text{and} \quad p = 0.868 > \alpha = 0.20$$

→ H0 is Not Rejected.
No significant difference across Easy (16.40), Medium (16.33) and Hard (16.27).

H4 — Two-Way ANOVA — Model × Language Interaction

Step 1 — Cell Means (n = 30 per cell)

Model	Python (X̄)	C++ (X̄)	Marginal Mean
Claude Sonnet 4.6	17.633	17.433	17.533
DeepSeek V3	16.867	16.367	16.617
Gemini 3 Flash	15.033	14.667	14.850
Marginal Mean	**16.511**	**16.156**	**16.333**

GM = 16.333 | a = 3 models | b = 2 languages | N = 180

Step 2 — SS(A) Effect of Model (n per marginal = 60)

$$SS(A) = \sum 60(\bar{X}_i - GM)^2$$

- Claude: $60 \times (17.533 - 16.333)^2 = 60 \times 1.440 = 86.400$
- DeepSeek: $60 \times (16.617 - 16.333)^2 = 60 \times 0.081 = 4.840$
- Gemini: $60 \times (14.850 - 16.333)^2 = 60 \times 2.199 = 131.993$

$$SS(A) = 86.400 + 4.840 + 131.993 = \boxed{223.233}$$

Step 3 — SS(B) Effect of Language (n per marginal = 90)

$$SS(B) = \sum 90(\bar{X}_j - GM)^2$$

- Python: $90 \times (16.511 - 16.333)^2 = 90 \times 0.0317 = 2.844$
- C++: $90 \times (16.156 - 16.333)^2 = 90 \times 0.0314 = 2.844$

$$SS(B) = 2.844 + 2.844 = \boxed{5.689}$$

Step 4 — SS(M) All Cells Combined (n per cell = 30)

$$SS(M) = \sum 30(\bar{X}_{ij} - GM)^2$$

- Claude/Python: $30 \times (17.633 - 16.333)^2 = 30 \times 1.690 = 50.700$
- Claude/C++: $30 \times (17.433 - 16.333)^2 = 30 \times 1.210 = 36.300$
- DeepSeek/Python: $30 \times (16.867 - 16.333)^2 = 30 \times 0.285 = 8.547$
- DeepSeek/C++: $30 \times (16.367 - 16.333)^2 = 30 \times 0.00116 = 0.034$
- Gemini/Python: $30 \times (15.033 - 16.333)^2 = 30 \times 1.690 = 50.700$
- Gemini/C++: $30 \times (14.667 - 16.333)^2 = 30 \times 2.778 = 83.334$

$$SS(M) = 50.700 + 36.300 + 8.547 + 0.034 + 50.700 + 83.334 = \boxed{229.600}$$

Step 5 — SS(A×B) Interaction

$SS(A \times B) = SS(M) - SS(A) - SS(B) = 229.600 - 223.233 - 5.689 = \boxed{0.678}$

Step 6 — SS(W) Within-Group Error

$$SS(W) = \sum(n_{ij} - 1) \times s_{ij}^2 = \boxed{104.400}$$

Step 7 — Degrees of Freedom

Source	df
Factor A (Model)	a−1 = 3−1 = **2**
Factor B (Language)	b−1 = 2−1 = **1**
Interaction A×B	(a−1)(b−1) = 2×1 = **2**
Within (error)	N−a · b = 180−6 = **174**
Total	N−1 = **179**

Step 8 — MS and F for Each Term

$$MS(A) = \frac{223.233}{2} = 111.617 \quad \Rightarrow \quad F_A = \frac{111.617}{0.600} = 186.028$$

$$MS(B) = \frac{5.689}{1} = 5.689 \quad \Rightarrow \quad F_B = \frac{5.689}{0.600} = 9.481$$

$$MS(A \times B) = \frac{0.678}{2} = 0.339 \quad \Rightarrow \quad F_{A \times B} = \frac{0.339}{0.600} = 0.565$$

$$MS(W) = \frac{104.400}{174} = 0.600$$

Step 9 — ANOVA Table

	SS	df	MS	F	Decision
Factor A (Model)	223.233	a−1 = 2	223.233/2 = 111.617	111.617/0.600 = 186.028	Tail area above F = < **0.001**
Factor B (Language)	5.689	b−1 = 1	5.689/1 = 5.689	5.689/0.600 = 9.481	Tail area above F = **0.002**
Interaction (A×B)	0.678	(a−1)(b−1) = 2	0.678/2 = 0.339	0.339/0.600 = 0.565	Tail area above F = **0.570**

	SS	df	MS	F	Decision
Within (error)	104.400	N−a · b = 174	104.400/174 = 0.600		
Total	333.999	N−1 = 179			

Step 10 — Decision

F critical values (= 0.20): F(2,174) = 1.624 · F(1,174) = 1.655

Term	F	F_crit	p-value	Decision
Model	186.028	1.624	< 0.001	**Reject H0**
Language	9.481	1.655	0.002	**Reject H0**
Interaction A×B	**0.565**	**1.624**	**0.570**	**→ H0 Not Rejected**

The interaction term has $p = 0.570 > \alpha = 0.20$. **The model ranking does not change between Python and C++.**

H5 — Two-Way ANOVA — Model × Complexity Interaction

Step 1 — Cell Means (n = 20 per cell)

Model	Easy ($\bar{X}$)	Medium ($\bar{X}$)	Hard ($\bar{X}$)	Marginal Mean
Claude Sonnet 4.6	17.850	17.350	17.400	17.533
DeepSeek V3	16.450	16.650	16.750	16.617
Gemini 3 Flash	14.900	15.000	14.650	14.850
Marginal Mean	**16.400**	**16.333**	**16.267**	**16.333**

GM = 16.333 | a = 3 models | c = 3 complexity levels | N = 180

Step 2 — SS(A) Effect of Model (n per marginal = 60)

$SS(A) = 60(17.533-16.333)^2+60(16.617-16.333)^2+60(14.850-16.333)^2 = \boxed{223.233}$

(same calculation as H1 and H4)

Step 3 — SS(C) Effect of Complexity (n per marginal = 60)

$$SS(C) = \sum 60(\bar{X}_c - GM)^2$$

- Easy: $60 \times (16.400 - 16.333)^2 = 60 \times 0.00449 = 0.267$
- Medium: $60 \times (16.333 - 16.333)^2 = 60 \times 0 = 0$
- Hard: $60 \times (16.267 - 16.333)^2 = 60 \times 0.00436 = 0.267$

$$SS(C) = 0.267 + 0 + 0.267 = \boxed{0.533}$$

Step 4 — SS(M) All Cells Combined (n per cell = 20)

$$SS(M) = \sum 20(\bar{X}_{ic} - GM)^2$$

- Claude/Easy: $20 \times (17.850 - 16.333)^2 = 20 \times 2.301 = 45.920$
- Claude/Medium: $20 \times (17.350 - 16.333)^2 = 20 \times 1.034 = 20.680$
- Claude/Hard: $20 \times (17.400 - 16.333)^2 = 20 \times 1.138 = 22.760$
- DeepSeek/Easy: $20 \times (16.450 - 16.333)^2 = 20 \times 0.0137 = 0.274$
- DeepSeek/Medium: $20 \times (16.650 - 16.333)^2 = 20 \times 0.1005 = 2.010$
- DeepSeek/Hard: $20 \times (16.750 - 16.333)^2 = 20 \times 0.1739 = 3.478$
- Gemini/Easy: $20 \times (14.900 - 16.333)^2 = 20 \times 2.054 = 41.080$
- Gemini/Medium: $20 \times (15.000 - 16.333)^2 = 20 \times 1.777 = 35.540$
- Gemini/Hard: $20 \times (14.650 - 16.333)^2 = 20 \times 2.832 = 56.640$

$SS(M) = 45.920+20.680+22.760+0.274+2.010+3.478+41.080+35.540+56.640 = \boxed{228.382}$

Step 5 — SS(A×C) Interaction

$SS(A \times C) = SS(M) - SS(A) - SS(C) = 228.382 - 223.233 - 0.533 = \boxed{4.616}$

Step 6 — SS(W) Within-Group Error

$$SS(W) = \sum (n_{ic} - 1) \times s_{ic}^2 = \boxed{105.500}$$

Step 7 — Degrees of Freedom

Source	df
Factor A (Model)	a–1 = 3–1 = **2**
Factor C (Complexity)	c–1 = 3–1 = **2**

Source	df
Interaction A×C	(a−1)(c−1) = 2×2 = **4**
Within (error)	N−a · c = 180−9 = **171**
Total	N−1 = **179**

Step 8 — MS and F for Each Term

$$MS(A) = \frac{223.233}{2} = 111.617 \quad \Rightarrow \quad F_A = \frac{111.617}{0.617} = 180.914$$

$$MS(C) = \frac{0.533}{2} = 0.267 \quad \Rightarrow \quad F_C = \frac{0.267}{0.617} = 0.432$$

$$MS(A \times C) = \frac{4.616}{4} = 1.154 \quad \Rightarrow \quad F_{A \times C} = \frac{1.154}{0.617} = 1.870$$

$$MS(W) = \frac{105.500}{171} = 0.617$$

Step 9 — ANOVA Table

	SS	df	MS	F	Decision
Factor A (Model)	223.233	a−1 = 2	223.233/2 = 111.617	111.617/0.617 = 180.914	Tail area above F = < **0.001**
Factor C (Complexity)	0.533	c−1 = 2	0.533/2 = 0.267	0.267/0.617 = 0.432	Tail area above F = **0.650**
Interaction (A×C)	4.616	(a−1)(c−1) = 4	4.616/4 = 1.154	1.154/0.617 = 1.870	Tail area above F = **0.118**
Within (error)	105.500	N−a · c = 171	105.500/171 = 0.617		
Total	333.882	N−1 = 179			

Step 10 — Decision

F critical values (= 0.20): F(2,171) = 1.625 · F(4,171) = 1.515

Term	F	F_crit	p-value	Decision
Model	180.914	1.625	< 0.001	**Reject H0**
Complexity	0.432	1.625	0.650	H0 Not Rejected
Interaction A×C	**1.870**	**1.515**	**0.118**	**→ Reject H0**

The interaction term has $p = 0.118 < \alpha = 0.20$. **The performance gap between models is not constant across complexity levels — H0 is Rejected.**

Summary Table — All Hypotheses (= 0.20)

H	Factor	F	df	p-value	F_crit (=0.20)	Decision
H1	Model (one-way)	178.358	(2, 177)	< 0.001	1.624	**Reject H0**
H2	Language (one-way)	3.084	(1, 178)	0.081	1.655	**Reject H0**
H3	Complexity (one-way)	0.142	(2, 177)	0.868	1.624	Not Reject H0
H4	Model × Language interaction	0.565	(2, 174)	0.570	1.624	Not Reject H0
H5	Model × Complexity interaction	1.870	(4, 171)	0.118	1.515	**Reject H0**